

5.1: Práctica: Diseño de controlador para un sistema de segundo orden.

Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$\begin{aligned} V_e(s) &= R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] \\ L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] &= R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs} \\ V_s(s) &= R I_2(s) + \frac{I_2(s)}{Cs} \end{aligned}$$

Procedimiento algebraico

$$\begin{aligned} V_e(s) &= (Ls + 2R) I_1(s) - (Ls + R) I_2(s) \\ I_1(s) &= \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s) \\ V_s(s) &= (R + \frac{1}{Cs}) I_2(s) = \frac{CRs + 1}{Cs} I_2(s) \\ V_e(s) &= [(Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} - (Ls + R)] I_2(s) \end{aligned}$$

Función de transferencia

$$\begin{aligned} \frac{V_s(s)}{V_e(s)} &= \frac{\frac{CRs + 1}{Cs} I_2(s)}{\frac{3CLR s^2 + (3CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)} \\ &= \frac{(CRs + 1)(Ls + 2R)}{3CLR s^2 + (3CR^2 + L)s + 2R} \end{aligned}$$

$$\frac{V_s(s)}{V_o(s)} = \frac{(LRs^2 + (CR^2 + L)s + R)}{3CLR s^2 + (3CR^2 + L)s + 2R}$$

Error en estado estacionario

$$e_{ss} = \left(1 - \frac{V_s}{V_c}\right) = \left(1 - \frac{CLRs^2 + (RR^2 + L)s + R}{3CLR s^2 + (3CR^2 + L)s + 2R}\right) = 1 - \frac{1}{2} = 0.5$$

Estabilidad

$$R = 100 \text{ k}\Omega$$

$$C = 470 \mu\text{F}$$

$$L = 390 \text{ mH}$$

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2ac}$$

$$s_2 = -427350427.3$$

$$s_1 = -8.511 \times 10^{-3}$$

$$a = 3CLR = 0.05499$$

$$b = 5CR^2 + L = 23500000$$

$$c = 2R = 200000$$

Respuesta: subamortiguado

Bosquejo

